

Report Analysis Questions

1. A security analyst performed a network scan on the target 192.168.1.10 using the Nmap tool. The report identified several open ports, including HTTP (80), SSH (22), and MySQL (3306), along with the operating system and service versions. Analyze the report and explain the security risks associated with the identified services. Discuss how an attacker could exploit these services and recommend suitable security measures to protect the target system.

Kali linux Commands:

```
nmap 192.168.1.10
```

```
nmap -sV 192.168.1.10
```

```
sudo nmap -O 192.168.1.10
```

```
sudo nmap -A 192.168.1.10
```

```
nmap -p 22,80,3306 -sV 192.168.1.10
```

```
sudo nmap -A 192.168.1.10 -oN nmap_report.txt
```

```
cat nmap_report.txt
```

Explanation:

nmap 192.168.1.10 – Scans the target IP address and finds open ports.

nmap -sV 192.168.1.10 – Shows the versions of the services running on the open ports.

sudo nmap -O 192.168.1.10 – Detects the operating system of the target.

sudo nmap -A 192.168.1.10 – Performs a detailed scan to find the OS, service versions, and other information.

nmap -p 22,80,3306 -sV 192.168.1.10 – Scans only ports 22, 80, and 3306 and shows their service versions.

sudo nmap -A 192.168.1.10 -oN nmap_report.txt – Saves the scan results in a file named nmap_report.txt.

cat nmap_report.txt – Displays the saved scan report.

2. A phishing awareness campaign was conducted for employees of ABC Technologies Pvt. Ltd. using the GoPhish tool. The report indicates that 40% of employees clicked the phishing link, 20% entered their login credentials, and several users downloaded a malicious attachment. Analyze the report and identify the security weaknesses demonstrated by the employees. Recommend suitable user awareness programs, email security controls, and organizational policies to reduce the risk of successful social engineering attacks.

Kali linux Commands:

```
cd ~/gophish
```

```
ls
```

```
chmod +x gophish
```

```
sudo ./gophish
```

```
https://127.0.0.1:3333
```

```
cat config.json
```

Explanation

cd ~/gophish – Opens the GoPhish installation folder.

chmod +x gophish – Gives permission to run the GoPhish application.

sudo ./gophish – Starts the GoPhish phishing server.

https://127.0.0.1:3333 – Opens the GoPhish Admin Dashboard in a web browser.

cat config.json – Displays the GoPhish configuration file.

3. A DNS assessment report generated using the Dig tool shows that the organization's DNS server permits a successful DNS Zone Transfer (AXFR), exposing internal hostnames, mail servers, and DNS records. Analyze the report and explain the security implications of the exposed information. Discuss how attackers could exploit these findings and recommend best practices to secure the DNS infrastructure.

Kali linux Commands:

```
dig example.com ANY
```

```
dig example.com NS
```

```
dig example.com MX
```

```
dig AXFR example.com @192.168.1.10
```

```
dig example.com A
```

```
dig example.com CNAME
```

```
dig AXFR example.com @192.168.1.10 > dns_report.txt
```

```
cat dns_report.txt
```

Explanation:

dig example.com ANY – Displays all available DNS records for the domain.

dig example.com NS – Displays the Name Server (NS) records of the domain.

dig example.com MX – Displays the Mail Exchange (MX) records used for email.

dig AXFR example.com @192.168.1.10 – Attempts a DNS Zone Transfer (AXFR) from the DNS server.

dig example.com A – Displays the IPv4 (A) record of the domain.

dig example.com CNAME – Displays the Canonical Name (CNAME) record of the domain.

dig AXFR example.com @192.168.1.10 > dns_report.txt – Saves the DNS Zone Transfer output to dns_report.txt.

cat dns_report.txt – Displays the saved DNS report.